

Hotel Chain Performance Analysis

Business insights report

Project overview – This Project Analyzes Hotel bookings Data to evaluate Occupancy, Revenue generation, Customer booking behavior, Cancellation trends, and operational performance. The objective is to identify actionable insights that support strategic business decision-making

Tool and techniques used

-  Microsoft Excel
-  Power BI
-  Power Query
-  Data Preprocessing
-  Data Modelling
-  DAX For Calculated Measures and Columns
-  Data Visualization
-  Dashboarding

Dataset overview

Data Set Name	Schema type	Total Columns	Total Records	Description
Dim_Date	Dimension Table	4	92	Dates of Analysis Period
Dim_Rooms	Dimension Table	2	4	Rooms and types
Dim_Hotels	Dimension Table	4	25	List of Hotels
Fact_aggregated_bookings	Fact Table	5	9200	Records of each hotel's bookings and capacity for every day
Fact_bookings	Fact Table	12	134,590	All the operational details such as bookings, revenue etc....

Metrics	Values
Total no of Months	3 (May, June, July)
Total no of Weeks	13 (19-31)
Total No of Days	91 Days
Total No of Cities	4 (Bangalore, Hyderabad, Delhi, Mumbai)
Total No of Hotels	25

Project workflow



Data collection

(Collected data from web sources)



Data preparation

(Preparing data suitable for analysis i.e. Data Cleaning.)



Data transformation

(Standardization and creation of new columns)



Data Modelling

(Relation Between Different tables)



DAX Calculations

(Using DAX formulas to create measures)



Dashboarding

(Built an interactive BI dashboard)



Business Insights

(Identified key trends and performance insights)



Recommendations

(Suggest actions based on analytical findings)

Dashboard overview



Total Bookings by Room Type

category

Business

Luxury

Elite

30.2K

18.7K

Standard

23.7K

14.3K

Premium

19.8K

10.4K

Presidential

9.0K

0K

Realisation% and ADR by booking platform

Realisation%

ADR(Avg Daily Rate)

logtrip

12705

70.57%

journey

12656

70.54%

direct online

12638

70.27%

others

12794

70.20%

makeouttrip

12702

70.09%

tripster

12778

69.80%

Key Metrics

Property ID	Property Name	city	Revenue	Total Bookings	RevPAR	Occupancy%	ADR	DSRN	DBRN	DURN	Realisation%	Cancellation	Avg Ratings
19558	Atliq Grands	Bangalore	54M	4316	5,527	44.33%	12,468	107.00	47.43	33.23	70.06%	1057	2.37
17564	Atliq Seasons	Mumbai	65M	3934	7,397	44.57%	16,597	97.00	43.23	30.52	70.59%	976	2.30
18559	Atliq Exotica	Hyderabad	47M	5192	4,061	44.57%	9,111	128.00	57.05	40.26	70.57%	1263	2.33
17562	Atliq Bay	Mumbai	51M	3388	6,803	44.86%	15,167	83.00	37.23	25.91	69.60%	862	2.37
18563	Atliq Palace	Hyderabad	44M	4669	5,014	52.89%	9,480	97.00	51.31	35.69	69.57%	1214	3.07
17560	Atliq City	Mumbai	87M	5940	7,763	53.07%	14,629	123.00	65.27	45.37	69.51%	1492	3.04
19561	Atliq Rili	Bangalore	72M	5669	6,774	53.25%	12,722	117.00	62.30	43.48	69.80%	1397	3.08
Total			1688M	132939	7,337	57.79%	12,696	2,528.00	1,460.87	1,024.64	70.14%	3302	3.62

RevPAR and ADR(Avg Daily Rate) by Month and Week num

RevPAR

ADR(Avg Daily Rate)

19

20

21

22

23

24

25

26

27

28

29

30

31

May

June

July

Fig.1 Dashboard

Key performance indicators (KPIs)

The dashboard summarizes the overall sales performance
Through key performance indicators (KPIs), helping stakeholders
Evaluate business performance briefly

*Note – All monetary values reported in this report are in **INR** unless stated otherwise, while large figures are displayed using international numbering convention (K, M, B)*

KPIs	Value	Description
Revenue	1.69 Billion	Total Revenue generated from successful bookings during analysis period
Occupancy	57.79%	Percentage of available rooms that were occupied. High occupancy indicates better room utilization.
ADR	12.70 k	Average Revenue earned per occupied room. Measures the hotel's pricing performance.
RevPAR	7.34 k	Revenue generated per available room, combining both occupancy and pricing efficiency.
Realisation	70.1%	Percentage of successful check-ins compared to total bookings after considering cancellations and no show, Indicates booking conversion efficiency.
DSRN	2.6 k	Average no of rooms available for sale each day across hotel portfolio, Represents total room inventory capacity
DBRN	1.46 k	Average no of rooms booked per day across the hotel portfolio

These KPIs represent the primary executive metrics selected for the dashboard.
Additional supporting measures were created in Power BI to enable detailed analysis, comparison, trends and business insights.

Business insights

The dashboard analysis revealed some important sales Trends, customer purchasing patterns and store performance Metrics. These insights can help management make informed Business decisions and improve overall operational efficiency.

Occupancy Analysis

Analysis area	Business interpretation
Occupancy (Weekday vs Weekend)	Occupancy is 62.64% in weekends while in weekdays it is 55.85% that reflects customer demand is stronger during weekends
Occupancy (Overall)	Overall Occupancy is 57.79% , Indicates that a substantial proportion of rooms inventory remains unsold during the analysis period.
Occupancy Vs ADR	Despite an approximate 8% increase in weekend occupancy, ADR remained relatively stable across weekdays and weekends. This suggests that room pricing was largely consistent despite variations in customer demand.
Occupancy Vs RevPAR	Across weeks 19-31, ADR remained relatively stable (approximately 12.6k – 12.7k), while RevPAR increased from 6.0k to 7.5k as occupancy improved from 50% to 60%. This indicates that, during the analyzed period, changes in RevPAR were primarily associated with improvement in occupancy rather than fluctuations in room pricing.
City Performance	Mumbai emerged as the strongest-performing market, generating the highest revenue (₹660.64M) and the highest RevPAR (₹8.90K) , supported by the highest ADR (₹15.38K) . In contrast, Delhi achieved the highest occupancy (60.44%) despite a comparatively lower ADR (₹12.16K), indicating strong customer demand. Bengaluru demonstrated balance revenue performance by maintaining the second-highest revenue (₹415.03M) with the second-highest ADR (₹13.13K), Hyderabad recorded the lowest ADR (₹9.32K) and RevPAR (₹5.40K) , despite occupancy (57.98%) and realisation (~70%) remaining comparable with other cities, suggesting relatively weaker revenue generation per available room.

Revenue by Room Category	Revenue efficiency increased consistently across higher room classes. The Presidential room class achieved the highest ADR (₹23.43K) and the highest RevPAR (₹13.86K) , despite having occupancy (59.14%) and realisation (~70.5%) levels comparable to other room classes. Conversely, the Standard room class recorded the lowest ADR (₹8.05K) and RevPAR (₹4.65K) . This indicates that higher room classes generated significantly greater revenue per available room primarily through premium pricing rather than higher occupancy.
Booking Conversion	The overall Realisation rate remained consistently around 70.14% across different business dimensions, including cities, months, and weeks . Even after applying multiple filters, only minor fluctuations of approximately 1-2 percentage points were observed, indicating a stable booking conversion pattern throughout the analysis period. However, a 70.14% Realisation rate also implies that nearly 30% of confirmed bookings did not result in actual check-ins, reducing the hotel's revenue realization and overall room utilization. This consistent trend across all dimensions suggests that booking losses were systemic rather than being concentrated in a particular city, month, or week.
Booking Channell performance	The Others booking channel generated the highest number of bookings and consequently contributed the highest share of revenue, while the Direct Offline channel recorded the lowest booking volume and revenue contribution. ADR (₹12.64K-₹12.79K) and realisation (69.80%-70.57%) remained largely consistent across all booking channels, indicating that variations in revenue were primarily driven by booking volume rather than pricing or booking conversion efficiency.

Key takeaways

- Weekend demand remained stronger than weekdays, with approximately **8% higher occupancy**.
- Overall occupancy remained at **57.79%**, indicating significant room inventory remained unutilized.
- Revenue Per Available Room was influenced more by **occupancy changes** than ADR during the analyzed period.
- **Mumbai** emerged as the strongest-performing market, while **Hyderabad** recorded the lowest Revenue Per Available Room.
- Higher room classes (**Premium** and **Presidential**) generated superior revenue efficiency through **higher ADR**, despite similar occupancy levels.
- The **Realisation rate remained consistently around 70%**, implying that nearly **30% of confirmed bookings** did not convert into actual check-ins across cities, weeks, and months.

Recommendations -:

Based on the findings identified during the business analysis, the following recommendations are proposed to improve revenue performance, optimize room utilization, strengthen booking conversion, and support data-driven decision-making. Each recommendation is directly linked to the corresponding business insight.

Business Findings	Recommendation
Weekend demand is significantly higher than weekday demand.	Introduce targeted weekday promotions and corporate packages to improve weekday occupancy and better utilize available room inventory during low-demand periods.
Overall occupancy remained at 57.79%, leaving a considerable portion of rooms unoccupied.	Implement demand forecasting and dynamic pricing strategies to improve room utilization while maintaining healthy ADR levels.
Revenue Per Available Room was primarily driven by occupancy rather than ADR.	Prioritize occupancy growth through demand-generation initiatives instead of relying solely on price adjustments, as higher occupancy showed a greater impact on revenue performance during the analysis period.
Mumbai outperformed other cities, while Hyderabad recorded the lowest Revenue Per Available Room.	Benchmark Mumbai's pricing and revenue management practices and evaluate whether similar strategies can be adapted for lower-performing cities such as Hyderabad after considering local market conditions.
Higher room classes (Premium and Presidential) generated substantially higher Revenue Per Available Room through stronger ADR.	Promote premium room upgrades using personalized offers and upselling opportunities to increase the average revenue generated per booking while preserving occupancy levels.

Realisation remained around 70%, indicating that nearly 30% of confirmed bookings did not result in completed stays.	Strengthen booking confirmation and cancellation management through automated reminders, flexible cancellation policies, and selective prepaid/non-refundable booking options to improve booking conversion and reduce revenue leakage.
Booking channel revenue differences were primarily driven by booking volume rather than pricing or booking conversion.	Focus marketing efforts on high-performing booking channels while evaluating the cost-effectiveness of low-performing channels to optimize distribution strategy and maximize return on acquisition spend.

Conclusion

- The analysis successfully evaluated the hotel's performance using key business metrics such as **Revenue, Occupancy, ADR, Revenue Per Available Room, and Realisation**.
- Customer demand was found to be **stronger on weekends**, while overall occupancy indicated opportunities to improve room utilization.
- Revenue performance was primarily influenced by **occupancy, pricing strategy, room class, and city-wise market performance**.
- **Mumbai** emerged as the strongest-performing market, whereas **Hyderabad** showed comparatively lower revenue efficiency.
- Higher room classes generated better revenue through **premium pricing**, while booking conversion remained consistently around **70%**, indicating opportunities to reduce revenue leakage.
- The recommendations provided in this report can help improve **occupancy, booking conversion, revenue generation, and overall operational efficiency**.
- Overall, this analysis demonstrates how **data-driven decision-making** can support strategic planning and enhance business performance in the hospitality industry.

Thank you

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For complete Power BI dashboard and project files visit:

GitHub - <https://github.com/aadityzz/Hospitality-Analysis-Project>

LinkedIn - <https://www.linkedin.com/in/adityatiwarix>